

Assure the campaign, not just the transaction.

APAR turns emerging GenAI payment threats into rail-
correct synthetic campaigns, tests layered defenses,
and preserves a human promotion boundary.

SYNTHETIC • OFFLINE • REPLAYABLE

Dylan Moraes • Competition submission

APAR
PAYMENT ASSURANCE

01 Overview

02 Scenario

03 Replay

04 Investigation

05 Defenses

06 Assurance

APAR ASSURANCE / 01 OVERVIEW

LOCAL SCORER VERIFIED

THREAT POSTURE - SELECTED GOLDEN PATH

Adaptive payment assurance.

Detect the campaign—not only the transaction—and intervene before authorized funds become irrecoverable.

Run verified replay →

Inspect scenario controls >

THREAT / APP-MULE

HIGH-RISK PATH

AI-personalized APP scam and mule campaign

CURATED DECISION FOOTPRINT

12 EVENTS

CHALLENGE 100.0% REVIEW 98.0% DECLINE 100.0%

01 02 03 04 05 06 07 08 09 10 11 12

EVENT 01 100.0% DECLINE HOLD

PERSUASION AUTHORIZED TRANSFER MULE DISPERSION

ATTEMPTED VALUE \$500.00 CAMPAIGN PAYMENTS 10 EVIDENCE CLASS Synthetic only

12 CURATED REPLAY EVENTS

3 PAYMENT CONTEXTS

14 LINKED ENTITIES

0.92 THREAT-SOURCE CONFIDENCE

RUNTIME POSTURE

Local scorer - verified

ensemble_with_graph

CAPABILITY DELTA

What changes with generative fraud

Bounded changes to personalization and iteration speed; no claim of autonomous settlement access.

01 / BASELINE

Conventional APP scam

Static scripts, broad targeting, manual iteration, and authorized push payment initiation by the victim.

02 / BOUNDED DELTA

Adaptive persuasion

✓ Synthetic personalization

✓ Faster message iteration

✓ Campaign-scale feedback

RAIL CONTEXT

One assurance layer, three contexts

C Card

CNP testing and synthetic refund context

CURATED CONTEXT

A A2A

Selected APP scam and mule golden path

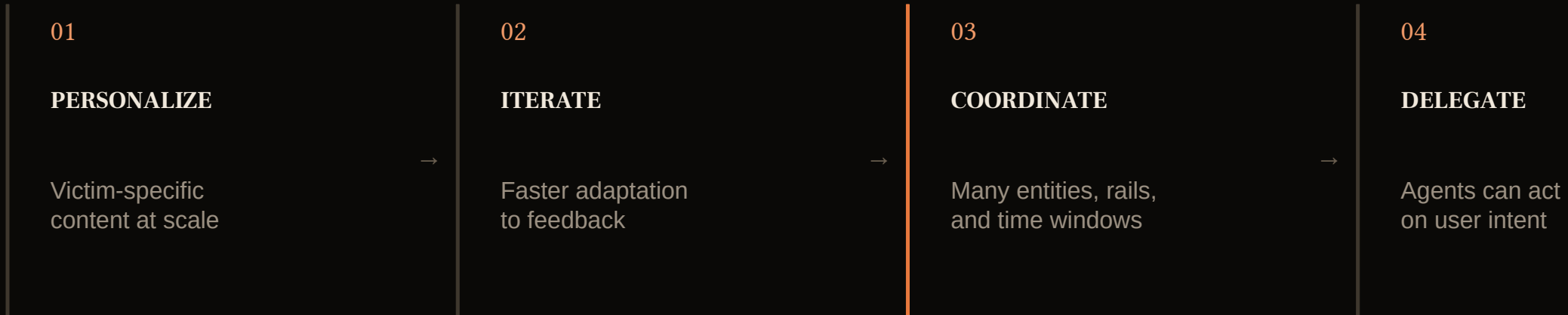
GOLDEN PATH

A Agentic

Intent and mandate-integrity context

CURATED CONTEXT

GenAI changes the operating model of fraud



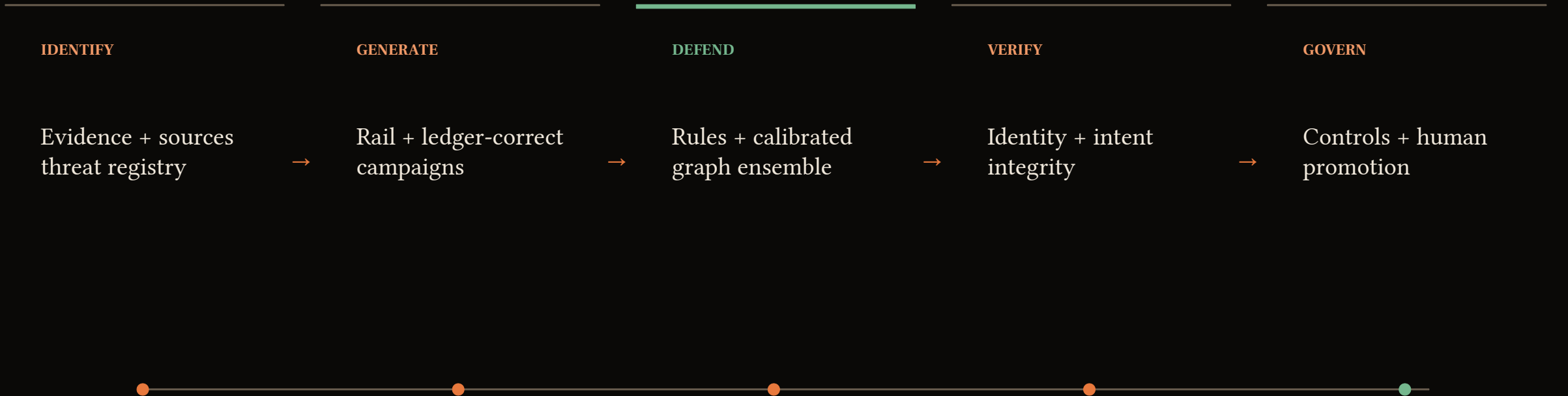
Transaction classifier



Campaign-aware assurance system

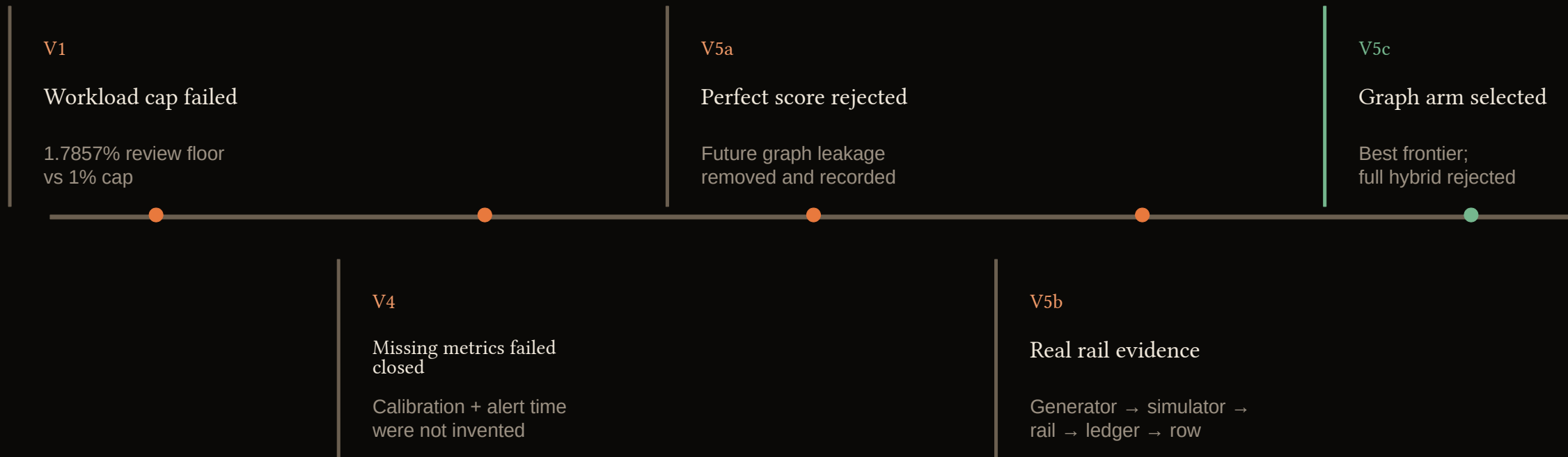
Behavior + lifecycle + economics + authority + governance

One assurance loop connects five control planes



Bounded feedback returns failed controls to the research backlog—never directly to deployment.

Failed experiments shaped the winning architecture



Novelty: not a single algorithm—an evidence-governed loop that learns from its own failure modes.

Graph context wins the operational frontier

Verified recovered diagnostics — synthetic and non-authoritative

ARM	RECALL %	PRECISION %	F1 %	FALSE-DECLINE %	P95 MS
Rules only	85.95	14.88	25.37	87.44	4.06
Ensemble • no graph	99.74	94.76	97.19	0.000	3.38
Ensemble • graph	99.87	95.88	97.83	0.0037	3.54
Full hybrid	99.93	16.85	28.83	87.44	19.74

Precision +1.12 pts

Challenge -0.23 pts

Latency +0.16 ms p95

A small, portable model powers the live demo

● Local scorer - verified
ensemble_with_graph

CONSUMER / 14

ATTACKER / 48

Organization 6
ORGANIZATION / 54

Beneficiary 7
BENEFICIARY / 10

\$50.00 CUMULATIVE ATTEMPTED

01 ● \$50.00

02 ● \$40.00

03 ● \$10.00

04 ● \$10.00

05 ● \$60.00

06 ● \$90.00

07 ● \$60.00

08 ● \$90.00

09 ● \$40.00

10 ● \$10.00

VICTIM TRANSFER

VICTIM TRANSFER

VICTIM TRANSFER

VICTIM TRANSFER

MULE LAYERING

CASH OUT

CASH OUT

VICTIM TRANSFER

VICTIM TRANSFER

VICTIM TRANSFER

14 ENTITIES

18 GENUINE GRAPH EDGES

SYNTHETIC ONLY

GRAPH 82172095-842437

INTERVENTION EXPLANATION

From calibrated probability to action.

REASON EVIDENCE

model_probability_decline_hold

The threshold ruler is bound to the selected portable event. It visualizes the model response only; campaign edges and post-event truth remain separate.

Latency was observed during this local scoring run and is not a production latency estimate.

TRUTH - SELECTED PORTABLE EVENT

WITHHELD FROM MODEL

Agentic Intent Abuse

OBSERVED LABEL

Positive fraud case

RAIL

AGENTIC

AMOUNT

\$10.00

REASON

model_probability_decline_hold

FEATURE VECTOR

46 bound features

EVENT 01 / 12

MANUAL TRACE INSPECTION BELOW

01	02	03	04	05	06
100.0%	100.0%	100.0%	70.5%	0.0%	1.2%
07	08	09	10	11	12
0.5%	17.7%	68.0%	100.0%	100.0%	100.0%

SCENARIO GRAPH

WITH

PORTABLE REPLAY

Shared presentation control - Independent evidence streams. No payment-to-trace record mapping asserted.

ORDERED PORTABLE TRACE

6 DECLINE HOLD 2 REVIEW HOLD 3 APPROVE 1 CHALLENGE

3 × CatBoost members

46 frozen causal features

12 hash-bound scenarios

0.0 max replay error

Approve • Challenge • Review hold • Decline hold

Investigators see one campaign—not twelve alerts

CASE-CENTRIC

14
entities

10
payment edges

\$500
conserved synthetic ledger

Analyst-time benefit: evidence pending

Victim 3
VICTIM · IN

Victim 4
VICTIM · SG

Consumer 5
CONSUMER · IN

Attacker 3
ATTACKER · SG

Organization 4
ORGANIZATION · IN

Attacker 5
ATTACKER · GB

Organization 6
ORGANIZATION · SG

Beneficiary 7
BENEFICIARY · IN

14 ENTITIES

10 PAYMENT EDGES

4 SYNTHETIC ILLICIT NODES

ARROW = PAYMENT DIRECTION

WEIGHT = AMOUNT

GRAPH: 82172095...842437

RUNTIME POSTURE

Local scorer - verified

ensemble_with_graph

FIRST CURATED APP INTERVENTION

Decline Hold
\$90.00 · event 02 · 100.0% calibrated

CUMULATIVE ATTEMPTED VALUE

\$500.00
10 ordered campaign payments

ANALYST TIME ESTIMATE

Evidence pending
No placeholder productivity claim

CASE GROUPING

1 campaign case
10 grouped events

VALUE PROGRESSION

Ordered campaign payments

LEDGER CONSERVED

#	TIME	STAGE	SOURCE → TARGET	AMOUNT	CUMULATIVE
01	00:04:51	Victim Transfer	980e05 → 39acaf	\$90.00	\$90.00
02	00:16:35	Victim Transfer	c5f474 → 39acaf	\$40.00	\$130.00

Agentic commerce needs proof before probability

- Agent identity

- Merchant + cart

- User mandate

- Expiry + nonce

- Scope + amount

- Replay rejection

DEFINITIVE FAILURE

DECLINE_HOLD before statistical risk

SEPARATE PROOF POINT • AGENTIC INTEGRITY

TrustVerifier rejects invalid authority.

Identity, mandate, scope, cart binding, and replay checks are genuine tested controls. This is a deterministic authorization proof—not evidence of graph-model performance.

TEST SOURCE

tests/trust/test_verifier.py

01	Identity AGENT_IDENTITY_MISMATCH	✓ TESTED
02	Mandate MANDATE_SCOPE_VIOLATION	✓ TESTED
03	Scope AMOUNT_LIMIT_EXCEEDED	✓ TESTED
04	Binding CART_HASH_MISMATCH	✓ TESTED
05	Replay NONCE_REPLAY	✓ TESTED

Evidence governance is part of the product

01

Causal features

Future append +
equal-time isolation

02

Executed controls

Shuffle, rename,
leakage, single-class

03

Immutable evidence

Models, traces,
metrics, manifests
hashed

04

Independent replay

Fresh environment re-
scores all 12 cases

05

Human promotion

The model cannot
approve itself

Portable bundle

52ed4c31...dbc900

Promotion boundary

Human approval required

NO SELF-PROMOTION

A sharp claim boundary makes the demo credible

WHAT WE PROVE

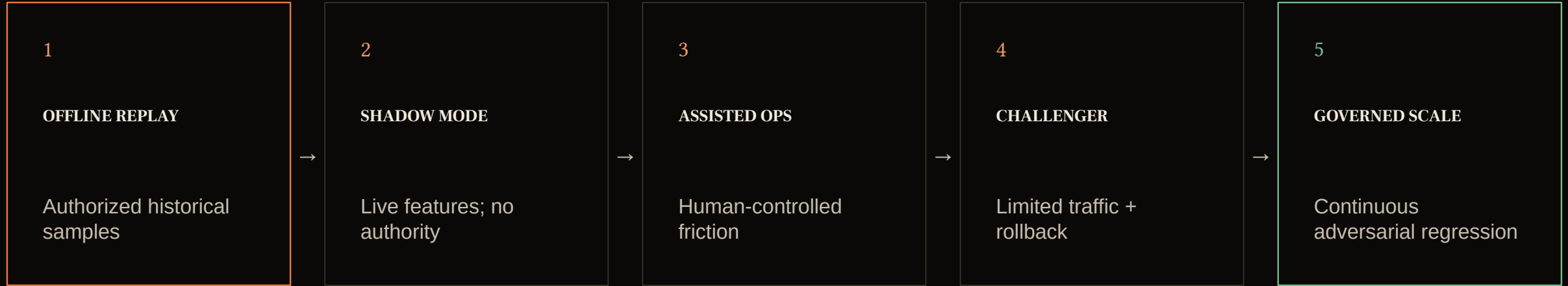
- ✓ Portable model loads and replays exactly
- ✓ Four rail-backed synthetic families
- ✓ Graph/no-graph architecture comparison
- ✓ Deterministic TrustVerifier integrity
- ✓ Offline console + hash-bound fallback

WHAT WE DO NOT CLAIM

- ✗ Production or real-cardholder performance
- ✗ Production-readiness or scale evidence
- ✗ External or multi-institution validation
- ✗ Autonomous model promotion
- ✗ Claims beyond the demonstrated model

Synthetic evidence before assertion.

From competition prototype to governed payment assurance



APAR

Assure the campaign. Verify the intent.
Promote only the evidence.

DEMO READY